

HOME LAB WAZUH

File Integrity Monitoring (FIM) + Integrasi VirusTotal dengan Active Response Otomatis

Dokumentasi laboratorium keamanan siber untuk demonstrasi deteksi perubahan file, analisis reputasi berbasis VirusTotal, dan respons otomatis penghapusan ancaman menggunakan platform open-source Wazuh.

Versi 1.2 · Wazuh 4.14.x · Target: Linux Agent · Portfolio / GitHub Ready

Tujuan & Capaian Pembelajaran

Setelah menyelesaikan lab ini, Anda diharapkan mampu:

1. Menginstal dan mengkonfigurasi Wazuh Manager + Dashboard menggunakan OVA VirtualBox.
2. Mendeploy Wazuh Agent pada host Linux dan memverifikasi koneksi ke manager.
3. Mengkonfigurasi File Integrity Monitoring (syscheck) secara real-time pada direktori target.
4. Mengintegrasikan VirusTotal API untuk analisis reputasi file yang terdeteksi FIM.
5. Membangun Active Response script yang otomatis menghapus file berbahaya.
6. Melakukan pengujian end-to-end menggunakan file uji EICAR dan memverifikasi alur deteksi → respons.

Skills yang Ditunjukkan

SIEM & Endpoint Detection | File Integrity Monitoring | Threat Intelligence Integration | Active Response / SOAR sederhana | Linux System Administration | Network Lab Design | Dokumentasi Teknis Profesional

Arsitektur Lab

Lab terdiri dari dua komponen utama yang berjalan di VirtualBox:

1. Wazuh Server (Manager + Indexer + Dashboard) — IP contoh : 192.168.100.119
2. Wazuh Agent (Linux) — Direktori target FIM: /home/wazuh/Downloads

Alur: Perubahan file → FIM → Alert → VirusTotal → Rule 87105 → Active Response → File dihapus.

Daftar Isi

1. Prasyarat dan Persiapan
2. Instalasi Server Wazuh (VirtualBox OVA)
3. Akses Dashboard dan Deployment Agent
4. Konfigurasi File Integrity Monitoring (FIM)
5. Integrasi VirusTotal
6. Active Response — Script Penghapusan Ancaman
7. Pengujian End-to-End
8. Troubleshooting
9. Catatan Keamanan dan Best Practice
10. Referensi

Prasyarat dan Persiapan

Pastikan lingkungan lab memenuhi persyaratan berikut sebelum memulai.

Perangkat Lunak

1. VirtualBox (versi terbaru) atau hypervisor lain yang mendukung format OVA
2. File OVA resmi Wazuh: wazuh-4.14.7.ova
3. Sistem operasi host stabil (Windows / Linux / macOS)
4. Koneksi internet

Sumber Daya Minimum

1. RAM VM Server: minimal 4 GB (disarankan 8 GB)
2. CPU: minimal 2 core · Storage: 20–30 GB
3. Satu VM Linux tambahan sebagai agent (Ubuntu/Debian)

Akun yang Diperlukan

Akun VirusTotal (gratis) → <https://www.virustotal.com> → Profile → API Key

Catatan: IP contoh 192.168.100.119. Sesuaikan dengan lab Anda.

Instalasi Server Wazuh (VirtualBox OVA)

Menggunakan OVA resmi adalah cara tercepat untuk mendapatkan Wazuh Manager, Indexer, dan Dashboard dalam satu paket.

Langkah Instalasi

1. Unduh file wazuh-4.14.7.ova dari situs resmi Wazuh.
2. VirtualBox → File → Import Appliance → pilih OVA.
3. Sesuaikan RAM/CPU jika perlu, lalu Import.
4. Start virtual machine.
5. Login konsol:

```
Username : wazuh-user
```

```
Password : wazuh
```

Peringatan: Segera ubah password default. Jangan gunakan kredensial default di lingkungan publik.

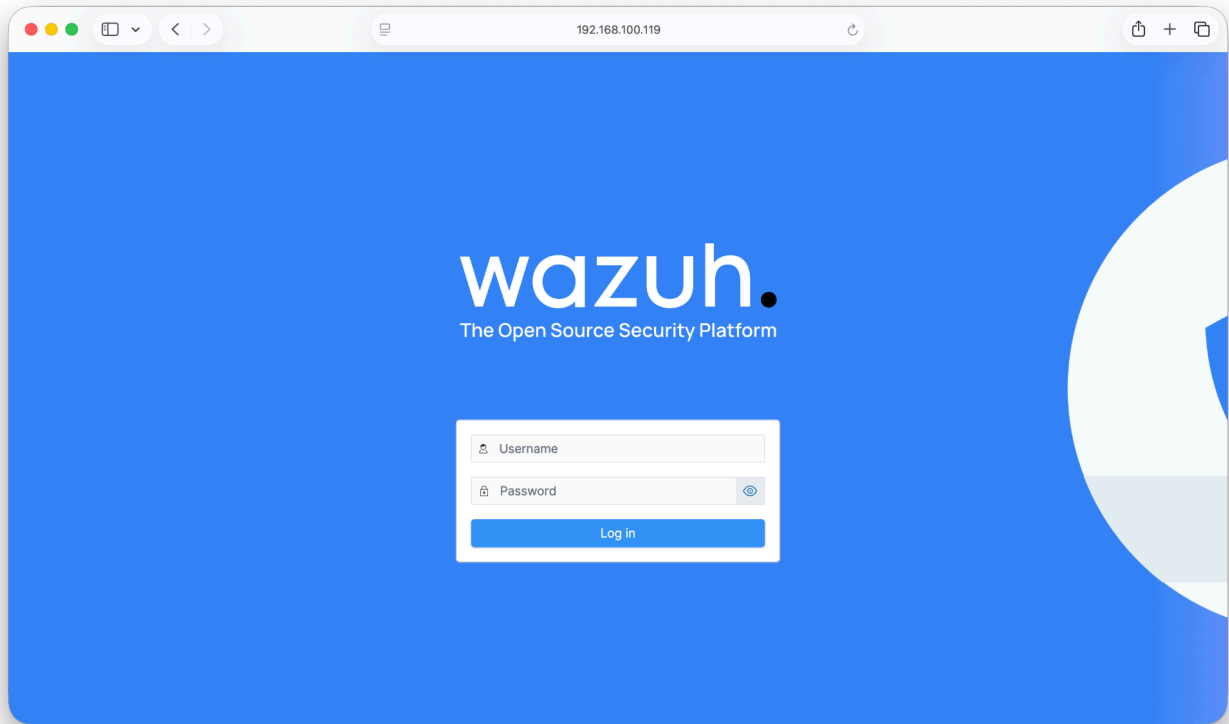
Akses Dashboard dan Deployment Agent

Login ke Wazuh Dashboard

Akses `http://192.168.100.119` (sesuaikan IP). Kredensial default:

Username : admin

Password : admin



Gambar 1. Halaman login Wazuh Dashboard

Setelah login, segera ubah password admin.

Deploy Agent Baru

1. Agents Management → Summary
2. Klik Deploy new agent

W.

Endpoints

Deploy new agent

API default a

Deploy new agent

LINUX

☒ RPM amd64
 ☐ RPM aarch64
 ☐ DEB amd64
 ☐ DEB aarch64

WINDOWS

☐ MSI 32/64 bits

macOS

☐ Intel
 ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

☒ Remember server address

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name

The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups

4 Run the following commands to download and install the agent:

```
curl -o wazuh-agent-4.14.7-1.x86_64.rpm https://packages.wazuh.com/4.x/yum/wazuh-agent-4.14.7-1.x86_64.rpm && sudo WAZUH_MANAGER='192.168.100.119' WAZUH_AGENT_NAME='ubntVM1' rpm -ihv wazuh-agent-4.14.7-1.x86_64.rpm
```

Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

5 Start the agent:

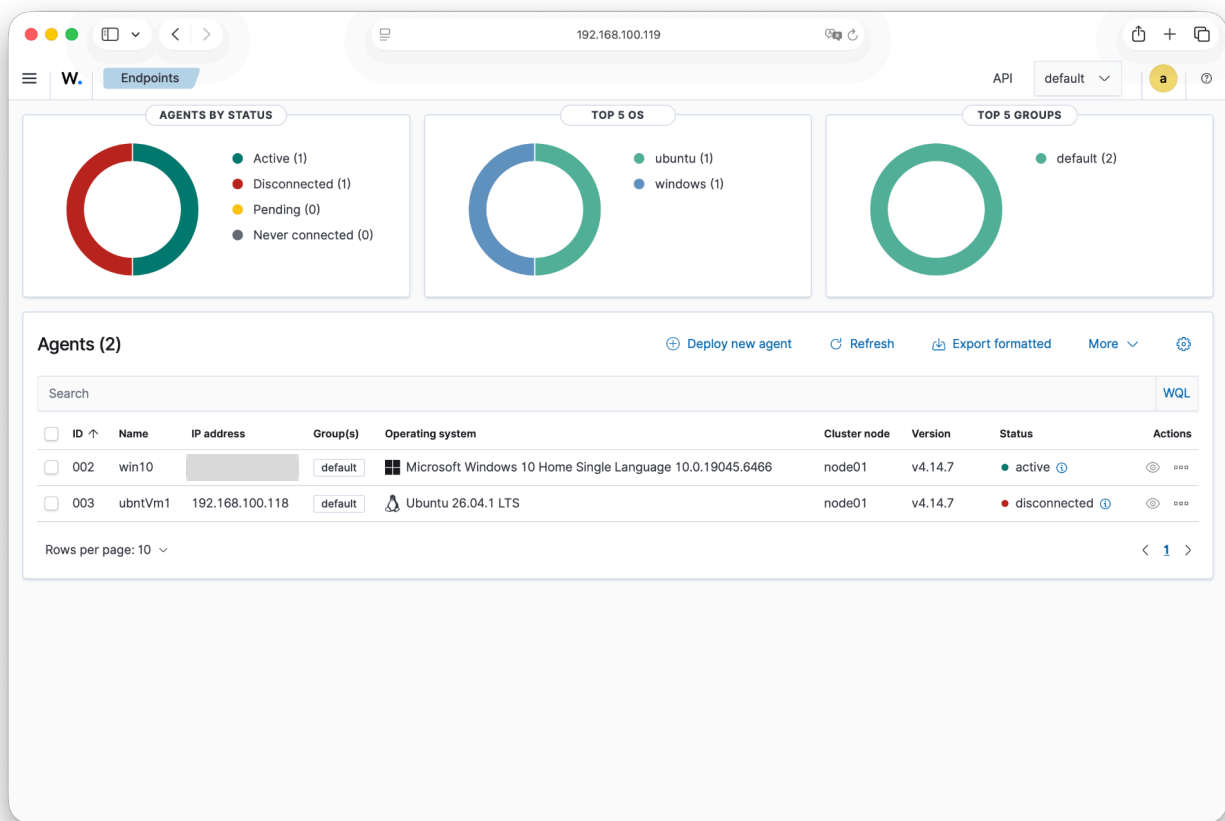
```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

6 Go to endpoints to verify the agent connection:

Back to agent list

Gambar 2. Form Deploy new agent

- Pilih OS target
- Isi Server address (IP Wazuh)
- Beri nama agent (contoh: agent01)
- Salin dan jalankan perintah instalasi di mesin agent (4)



Gambar 3. Ringkasan Agents

Aktifkan service agent:

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

Pastikan status agent Active di dashboard.

Konfigurasi File Integrity Monitoring (FIM)

Modul syscheck memantau perubahan file/direktori (create, modify, delete) secara real-time maupun terjadwal.

Siapkan Direktori Target

```
mkdir -p /home/wazuh/Downloads  
chmod 777 /home/wazuh/Downloads
```



Gambar 4. Membuat direktori Downloads

Permission 777 hanya untuk lab. Di produksi gunakan permission lebih ketat.

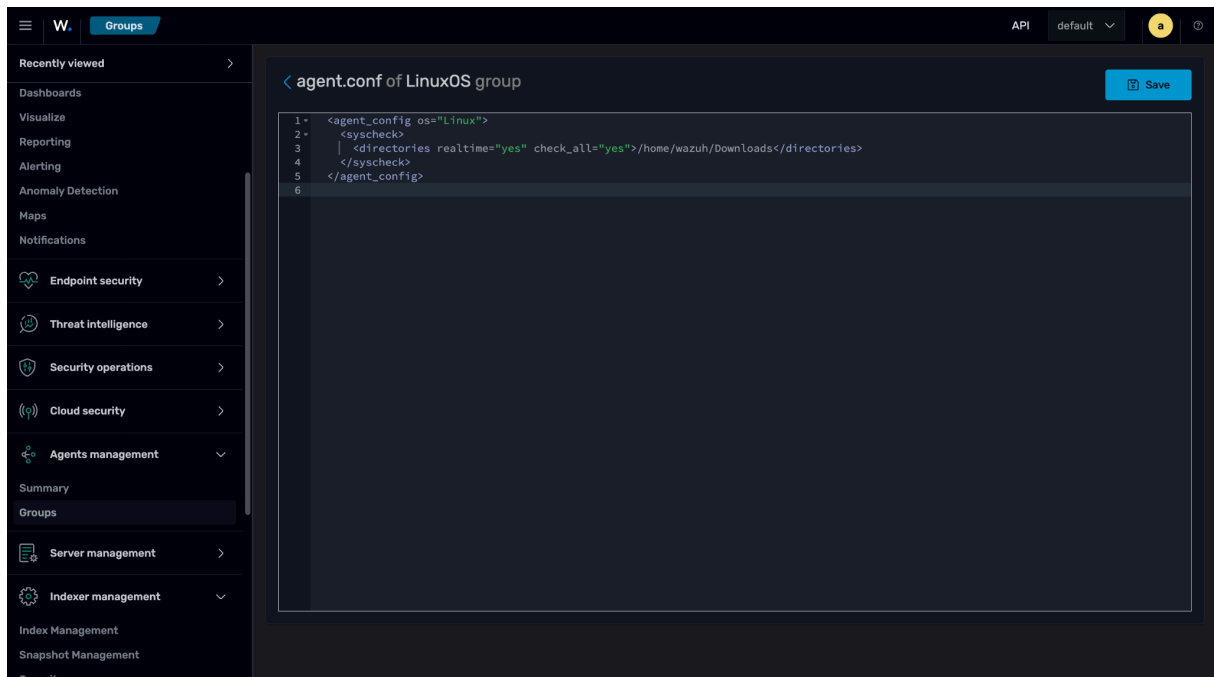
Konfigurasi melalui Agent Group

1. Agents Management → Groups → buat group LinuxOS

2. Edit agent.conf group:

```
<agent_config os="Linux">
  <syscheck>
    <directories realtime="yes" check_all="yes">/home/wazuh/Downloads</directories>
  </syscheck>
</agent_config>
```

realtime="yes" = real-time (inotify). check_all="yes" = cek ukuran, hash, permission, dll.



Gambar 5. Konfigurasi Agent Group

Konfigurasi Langsung di Agent

```
sudo nano /var/ossec/etc/ossec.conf
```

```
<syscheck>
  <disabled>no</disabled>
  <alert_new_files>yes</alert_new_files>
  <directories realtime="yes" check_all="yes">/home/wazuh/Downloads</directories>
</syscheck>
```

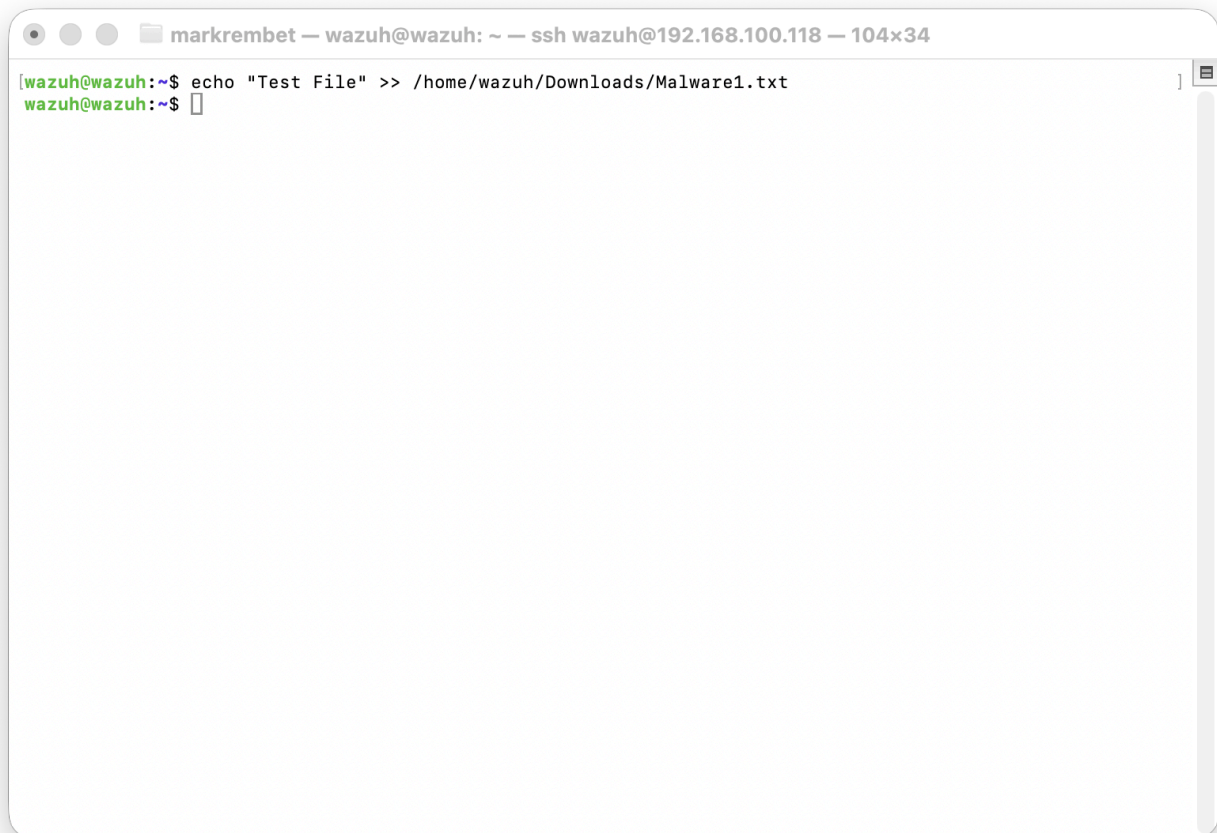
```
sudo systemctl restart wazuh-agent
```

Tekan ctrl + o lalu ctrl + x untuk menyimpan dan keluar dari nano.

Verifikasi FIM

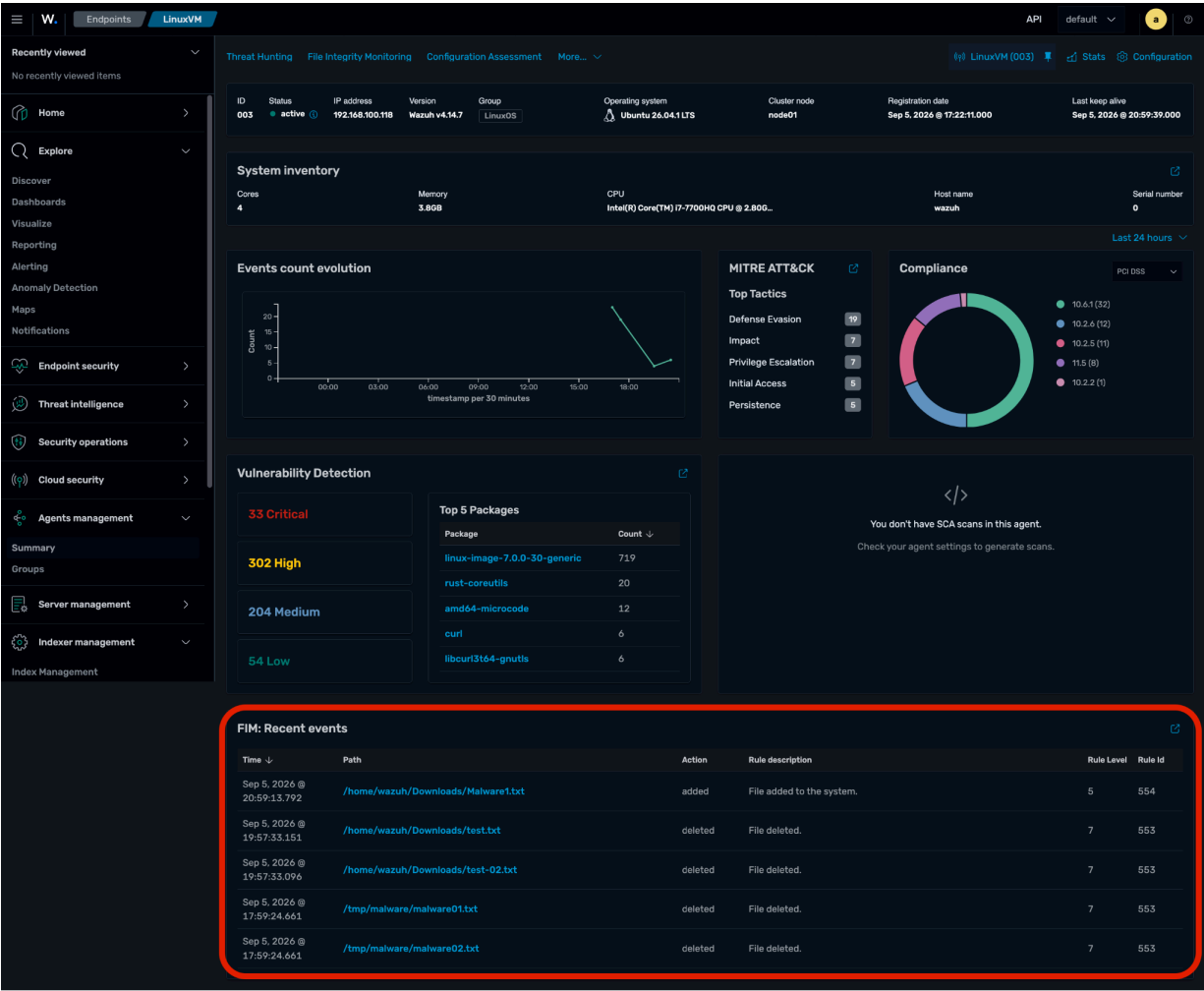
Wazuh agent terminal

```
echo "Test File" >> /home/wazuh/Downloads/Malware1.txt
```



Gambar 6. Membuat file uji FIM

Periksa File Integrity Monitoring → Events. Event "File added" harus muncul.

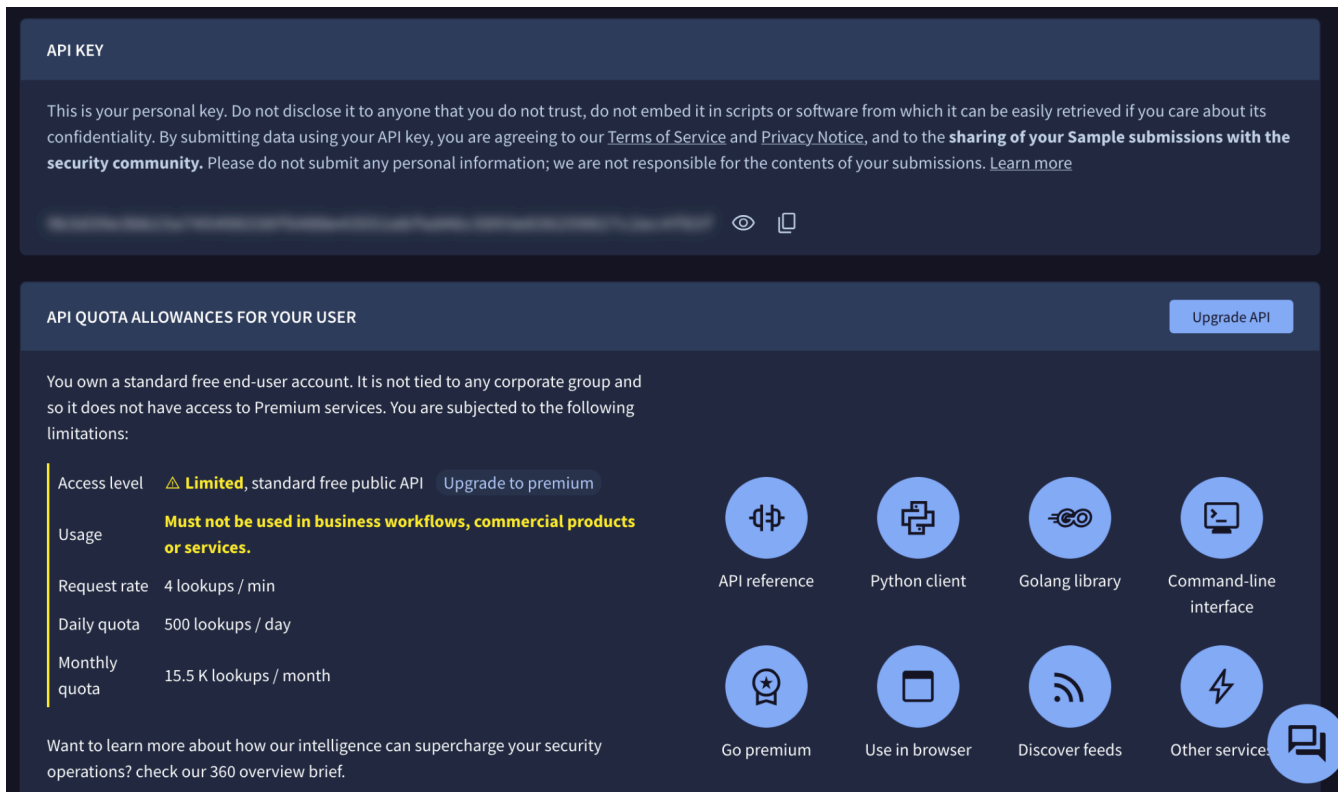


Gambar 7. Event FIM (added / modified / deleted)

Integrasi VirusTotal

Hash file yang terdeteksi FIM dikirim ke VirusTotal. Jika berbahaya, alert muncul dan dapat memicu Active Response.

Dapatkan API Key



API KEY

This is your personal key. Do not disclose it to anyone that you do not trust, do not embed it in scripts or software from which it can be easily retrieved if you care about its confidentiality. By submitting data using your API key, you are agreeing to our [Terms of Service](#) and [Privacy Notice](#), and to the **sharing of your Sample submissions with the security community**. Please do not submit any personal information; we are not responsible for the contents of your submissions. [Learn more](#)

API QUOTA ALLOWANCES FOR YOUR USER Upgrade API

You own a standard free end-user account. It is not tied to any corporate group and so it does not have access to Premium services. You are subjected to the following limitations:

Access level	⚠ Limited , standard free public API Upgrade to premium
Usage	Must not be used in business workflows, commercial products or services.
Request rate	4 lookups / min
Daily quota	500 lookups / day
Monthly quota	15.5 K lookups / month

Want to learn more about how our intelligence can supercharge your security operations? check our [360 overview brief](#).

Integration options:

- API reference
- Python client
- Golang library
- Command-line interface
- Go premium
- Use in browser
- Discover feeds
- Other service

Gambar 8. Virustotal API

1. Daftar/login di <https://www.virustotal.com>
2. Profile → API Key → salin key

Konfigurasi di Wazuh Manager

```
sudo nano /var/ossec/etc/ossec.conf

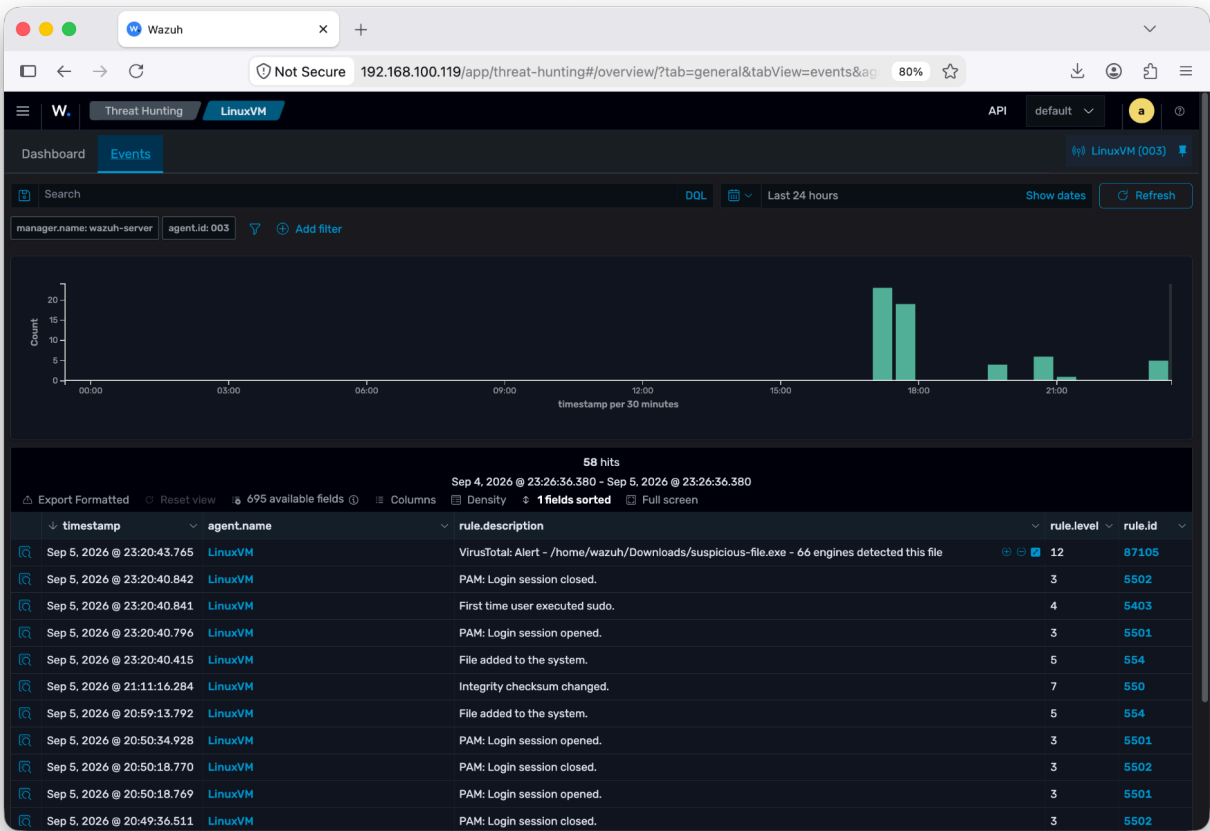
<integration>
  <name>virustotal</name>
  <api_key>API_KEY</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>

sudo systemctl restart wazuh-manager
```

Uji dengan EICAR

Wazuh agent terminal

```
sudo curl -Lo /home/wazuh/Downloads/suspicious-file.exe \
https://secure.eicar.org/eicar.com
```



Gambar 9. Alert VirusTotal di Threat Hunting

Active Response - Script Penghapusan Ancaman

Active Response menjalankan aksi otomatis ketika rule tertentu terpicu.

Script di Agent

```
sudo apt update && sudo apt install -y jq  
cd /var/ossec/active-response/bin
```

```
sudo nano remove-threat.sh  
  
#!/bin/bash  
LOCAL=`dirname $0`  
cd $LOCAL  
cd ../  
PWD=`pwd`  
LOG_FILE="${PWD}/logs/active-responses.log"  
  
read INPUT_JSON  
FILENAME=$(echo $INPUT_JSON | jq -r .parameters.alert.data.virustotal.source.file)  
COMMAND=$(echo $INPUT_JSON | jq -r .command)  
  
if [ "$COMMAND" == "add" ] && [ -f "$FILENAME" ]; then  
    rm -f "$FILENAME"  
    if [ $? -eq 0 ]; then  
        echo "`date '+%Y/%m/%d %H:%M:%S'` $0: Successfully removed threat" >> ${LOG_FILE}  
    else  
        echo "`date '+%Y/%m/%d %H:%M:%S'` $0: Error removing threat" >> ${LOG_FILE}  
    fi  
fi  
exit 0
```

```
sudo chmod 750 remove-threat.sh  
sudo chown root:wazuh remove-threat.sh  
sudo systemctl restart wazuh-agent
```

Trigger script virustotal pada wazuh-server file terinfeksi virus

```
sudo nano /var/ossec/etc/ossec.conf
```

```
<command>  
  <name>remove-threat</name>  
  <executable>remove-threat.sh</executable>  
  <timeout_allowed>no</timeout_allowed>  
</command>
```

```
<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>defined-agent</location>
  <agent_id>003</agent_id>
  <rules_id>87105</rules_id>
</active-response>
```

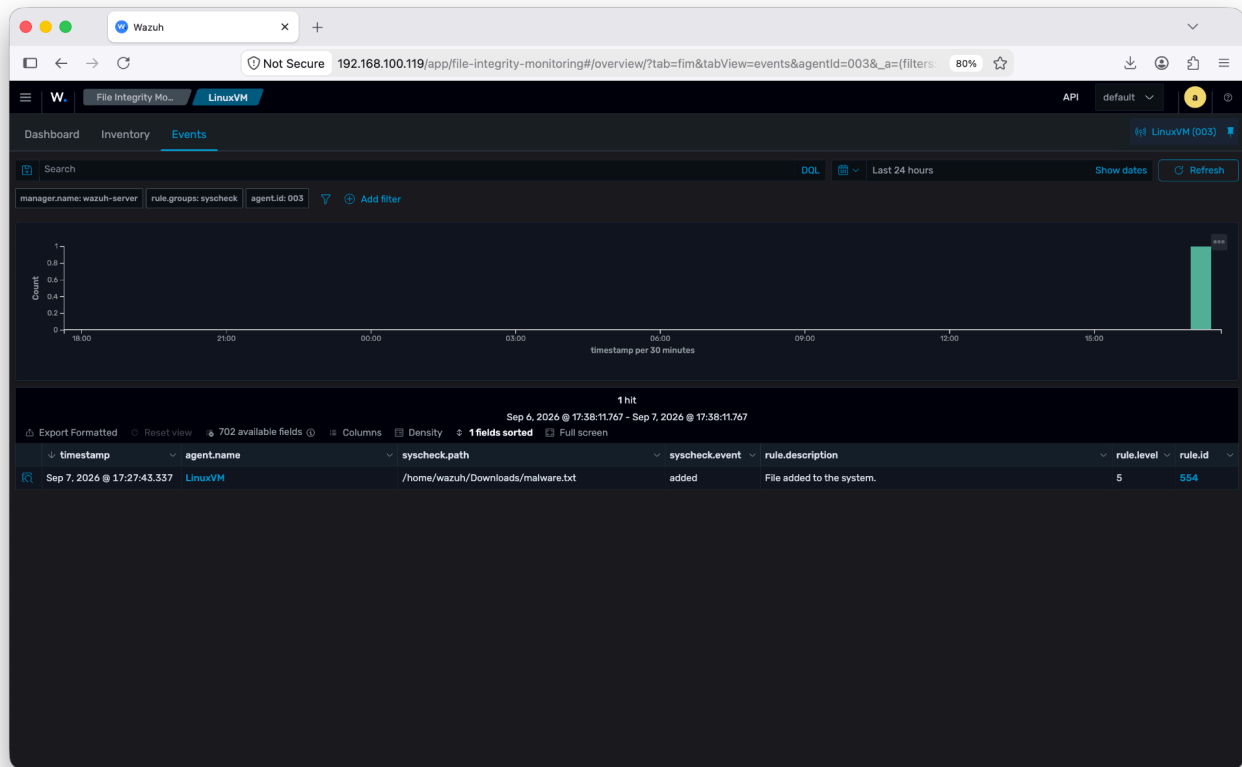
Sesuaikan agent_id. Rule 87105 = deteksi VirusTotal positif.

```
sudo systemctl restart wazuh-manager
```

Pengujian End-to-End

Uji File Bersih

```
echo "Testing" >> /home/wazuh/Downloads/malware.txt
```

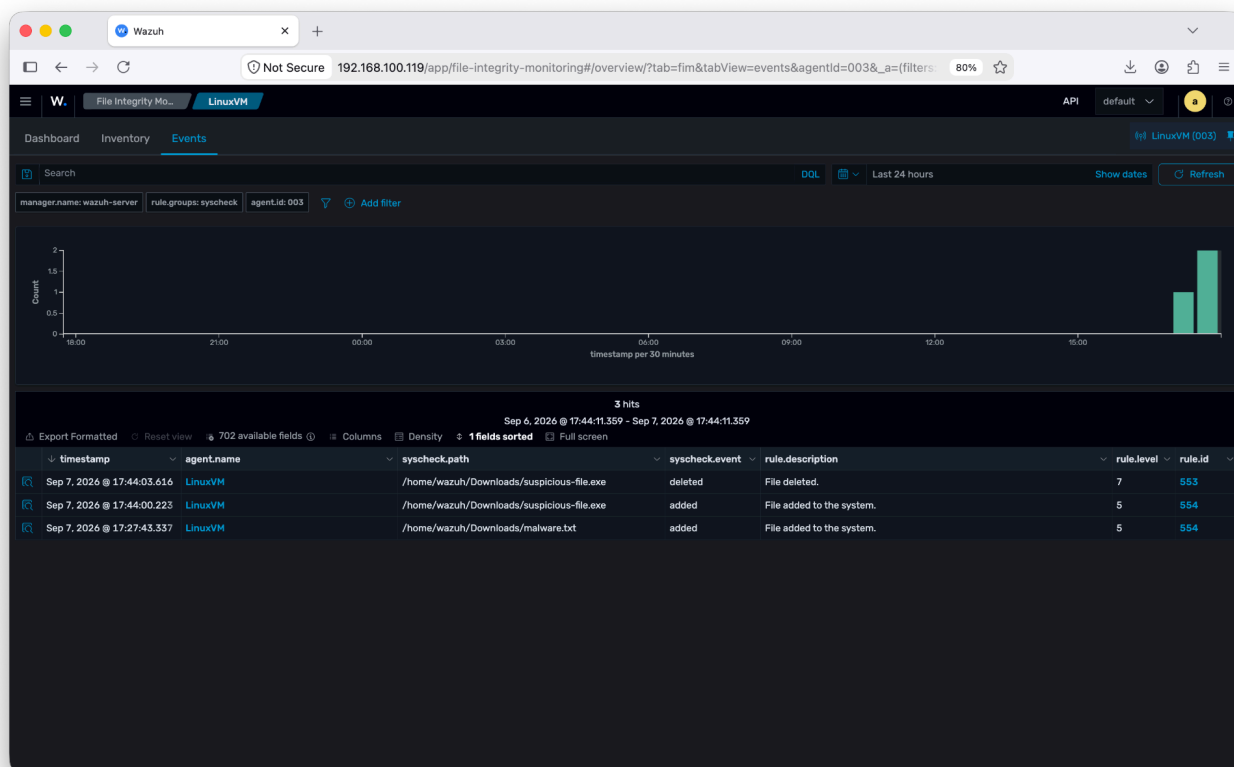


Gambar 10. File teks biasa (tidak berbahaya)

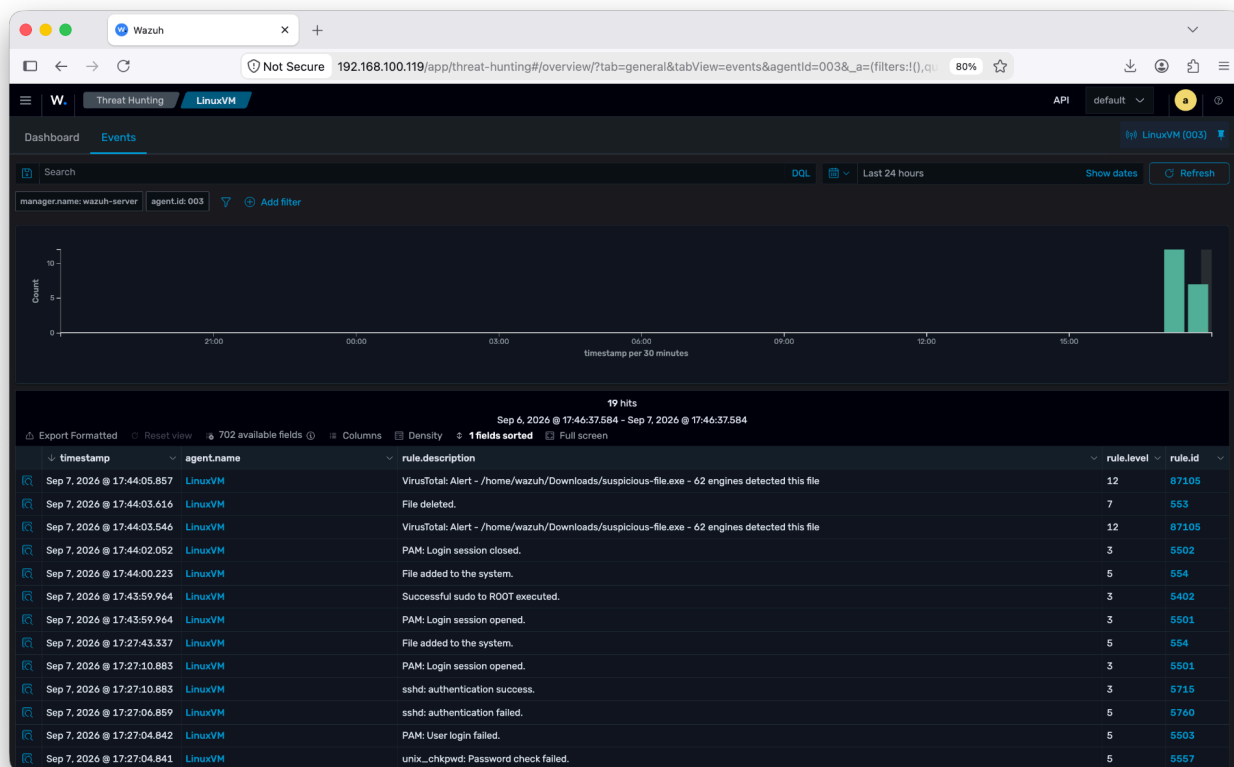
FIM mendeteksi file, VirusTotal tidak menandai berbahaya → Active Response tidak dijalankan.

Uji File Berbahaya (EICAR)

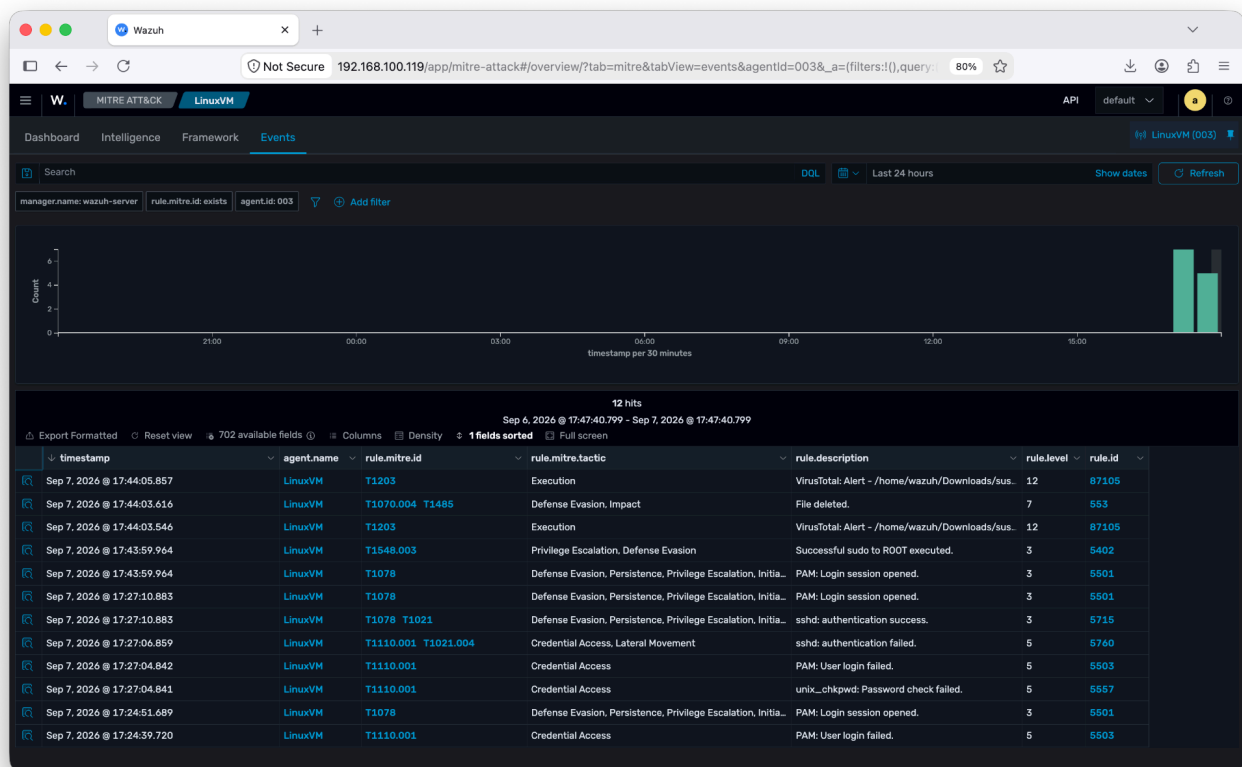
```
sudo curl -Lo /home/wazuh/Downloads/suspicious-file.exe \ https://secure.eicar.org/eicar.com
```



Gambar 11. Mengunduh file uji EICAR - Event FIM



Gambar 12. Mengunduh file uji EICAR - Event Threat Hunting



Gambar 13. Mengunduh file uji EICAR - Event MITRE ATT&CK

Alur yang diharapkan:

1. FIM → event File added
2. VirusTotal → alert positif (rule 87105)
3. Active Response → script menghapus file
4. FIM → event File deleted

Troubleshooting

1. Agent Disconnected — Cek IP manager, firewall (1514/tcp), status wazuh-agent
2. FIM tidak muncul event — Cek realtime=yes, path direktori, restart agent
3. VirusTotal tidak alert — Cek API key, rate limit, posisi integrasi di ossec.conf
4. Active Response tidak jalan — Cek agent_id, rules_id, permission script (750), ownership root:wazuh
5. Script error — Pastikan jq terinstal dan path file di JSON valid

Catatan Keamanan dan Best Practice

1. Segera ubah semua password default.
2. Jangan gunakan permission 777 di produksi.
3. Batasi akses dashboard (firewall / VPN).
4. Jangan commit API key VirusTotal ke repositori publik.
5. Active Response penghapusan file berpotensi berbahaya jika rule longgar.
6. Gunakan EICAR untuk pengujian; hindari malware nyata kecuali lingkungan terisolasi.
7. Backup konfigurasi sebelum perubahan besar.
8. Pantau log manager dan agent secara berkala.

Peringatan: Lab ini untuk pembelajaran dan portfolio. Produksi memerlukan hardening lebih lanjut.

Referensi

1. Dokumentasi resmi Wazuh — <https://documentation.wazuh.com>
2. VirusTotal API — <https://docs.virustotal.com>
3. EICAR Test File — <https://www.eicar.org>
4. Wazuh GitHub — <https://github.com/wazuh/wazuh>

Dokumen ini disusun sebagai portfolio teknis dan dipublikasikan di GitHub.

Happy learning & stay secure.